TripCraft: A Benchmark for Spatio-Temporally Fine-Grained Travel Planning

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Abstract

Existing travel-planning benchmarks rely on semi-synthetic data, exhibit spatial inconsistencies, and omit critical real-world constraints such as public transit schedules, event calendars, and fine-grained attraction types. Their evaluations are largely binary and fail to capture the temporal, spatial, and personalization quality of generated itineraries, limiting real-world applicability. We introduce *TripCraft*, a transit-aware, persona-conditioned benchmark grounded in real-world, constraint-rich data. *TripCraft* contains 1,000 travel queries across 140 U.S. cities for 3-, 5-, and 7-day itineraries at three difficulty levels, with human-annotated gold plans and rich user personas. Beyond conventional constraint pass-rate checks, *TripCraft* introduces five continuous evaluation metrics capturing temporal meal scheduling, temporal attraction pacing, spatial transit proximity, PoI ordering, and persona alignment. We demonstrate that parameter-informed prompting substantially improves itinerary quality under these metrics—with 7-day Temporal Meal Score rising from 0.61 to 0.80—while also exposing a feasibility trade-off in which certain constraint pass rates decline. *TripCraft* sets a new standard for evaluating LLM-based travel planning by combining real-world data with nuanced, multi-faceted metrics that reflect practical, personalized itinerary quality.

1 Introduction

Planning is a fundamental cognitive task (Kambhampati et al., 2023) and a long-standing challenge in artificial intelligence (Valmeekam et al., 2023; Bohnet et al., 2024). Travel planning is a particularly demanding instantiation: it requires jointly satisfying hard constraints (budget, event schedules, transportation logistics) and soft preferences (traveler persona, pace, location type), while adhering to temporal and geographic coherence across multiple days and cities.

Large language models (LLMs) have shown remarkable breadth across planning tasks (Wei et al., 2022; Yao et al., 2023; Xi et al., 2025), but assessing their travel-planning capabilities requires a benchmark that reflects the full complexity of real-world itinerary generation. The seminal TravelPlanner benchmark (Xie et al., 2024) was a crucial step forward, providing a rich sandbox environment with nearly four million data records. Its extension TravelPlanner+ (Singh et al., 2024) introduced personalization through user models. Despite these advances, both benchmarks rely on semi-synthetic or partially curated data and evaluate plans predominantly through binary constraint pass rates—a plan either satisfies all constraints or it does not. Such coarse metrics cannot distinguish a near-perfect itinerary from a fundamentally flawed one along continuous quality dimensions such as temporal scheduling realism, spatial transit accessibility, or persona fit.

Furthermore, existing benchmarks omit several practical constraints that are decisive in real-world travel: public-transit reachability (General Transit Feed Specification, or GTFS data), structured event categories (sports, arts, music, film), fine-grained attraction types with category-appropriate visit durations, and traveler personas that encode spending style and location preferences. Without these, an evaluation cannot distinguish a plan that schedules dinner at 2 p.m. from one aligned with natural meal timing, or one that places a user in a transit desert from one with convenient public transport access.

To address these gaps, we introduce ***TripCraft***, a **S**patio-**T**emporally **F**ine-**G**rained travel-planning benchmark. *TripCraft* makes the following contributions:

- **A large-scale, realistic benchmark** of 1,000 queries across 140 U.S. cities, spanning 3-, 5-, and 7-day itineraries at three difficulty levels, backed by real-world data including GTFS

public transit, live event calendars, typed attraction categories, and flight records.

- **Rich user personas** encoding traveler type (laidback vs. adventure seeker), purpose of travel, spending preference, and location preference, each injected into both the query and the evaluation.

- **Human-annotated gold itineraries** produced by 25 graduate annotators with rationales, refined through multi-round expert review, and used to derive per-day evaluation parameters from annotation statistics.

- **Five novel continuous evaluation metrics**: Temporal Meal Score ($\bar{T}_{meal}$), Temporal Attraction Score ($\bar{T}_{attrac}$), Spatial Score ($\bar{S}_{spatial}$), Ordering Score ($\bar{S}_{ord}$), and Persona Score ($\bar{S}_{persona}$), complementing existing binary constraint checks with nuanced quality signals.

- **Empirical evidence** that parameter-informed prompting substantially improves itinerary quality under the proposed metrics while revealing a tension between itinerary realism and strict constraint satisfaction.

2 Related Work

Travel planning benchmarks. TravelPlanner (Xie et al., 2024) introduced a challenging real-world planning benchmark using flights, hotels, and attractions, showing that even GPT-4 achieves only 0.6% success under strict constraint satisfaction. TravelPlanner+ (Singh et al., 2024) extended this with personalization via user models and LLM-as-a-judge evaluation. ChinaTravel (Shao et al., 2024) provides a Chinese-language counterpart with country-specific constraints. TravelAgent (Chen et al., 2024) focuses on system design for personalized travel planning with tool-use, recommendation, and memory modules. *TripCraft* differs from these in: (1) grounding all data in real-world sources including GTFS transit and live events, (2) introducing persona-conditioned queries, and (3) providing five continuous evaluation metrics alongside standard constraint checks.

LLM planning capabilities. A growing body of work investigates whether LLMs can autonomously plan (Valmeekam et al., 2023; Bohnet et al., 2024; Yang and Tomar, 2023; Pallagani et al., 2023). Chain-of-thought (Wei et al., 2022) and ReAct (Yao et al., 2023) prompting have improved multi-step reasoning; the LLM-Modulo framework (Gundawar et al., 2024) and formal verification tools (Hao et al., 2025) have pushed constraint satisfaction rates on TravelPlanner. Mind Evolution (Lee et al., 2025) achieves over 98% solution rate on TravelPlanner using evolutionary search at inference time. ADaPT (Prasad et al., 2024) addresses task decomposition for embodied agents. Despite these advances, the field lacks evaluation frameworks that capture the continuous, multi-dimensional quality of travel plans beyond binary pass rates, which *TripCraft* addresses.

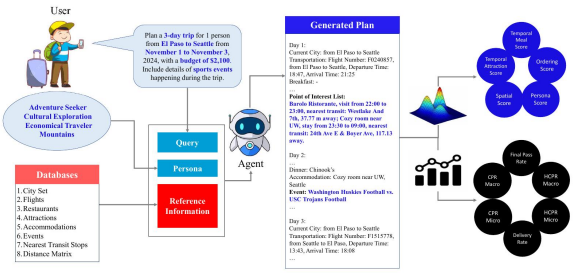

Figure 1: *TripCraft* overview: continuous evaluation metrics alongside fine-grained itinerary details including temporal scheduling, spatial transit proximity, PoI sequencing, and persona alignment.

Spatial and transit-aware data. Point-of-interest (PoI) data integration and conflation (Sun et al., 2023) is a recognized challenge. Recent work has explored LLM integration with GTFS transit data (Jonnala et al., 2025). *TripCraft* systematically integrates GTFS data for 140 cities to evaluate spatial transit proximity as a core quality dimension.

3 *TripCraft* Benchmark

3.1 Overview

Figure 1 illustrates the *TripCraft* pipeline. The benchmark contains 1,000 travel queries spanning three itinerary lengths (3-day, 5-day, 7-day) and three difficulty levels (Easy, Medium, Hard) across 140 U.S. cities. Each query is paired with a user persona and a human-annotated gold itinerary. Evaluation combines traditional constraint checking with five novel continuous metrics parameterized from annotation statistics.

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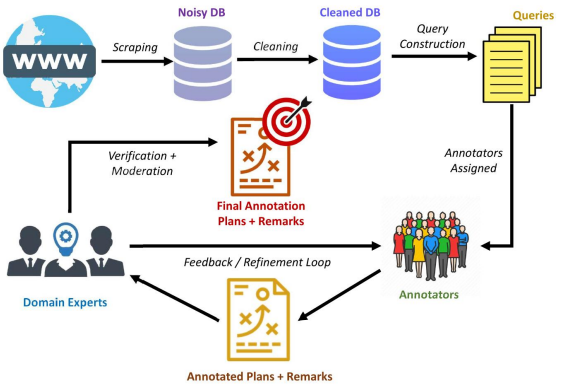

Figure 2: Visualization of the *TripCraft* construction pipeline, from raw data collection through query generation, annotation, and metric parameterization.

3.2 Data Construction Pipeline

Figure 2 shows the data construction pipeline. The database spans multiple real-world sources, as summarized in Table 1.

Database	Data Entries (#)
City Set	140
Flights	3,446,829
Restaurants	3,892
Attractions	5,043
Accommodations	2,800
Events	21,980
Nearest Transit Stop	8,723
Distance Matrix	19,460

Table 1: Data entries in the database have been scraped to include the most recent data.

Data sources. Attractions (5,043 entries) and restaurants (3,892) were collected via web scraping and OpenStreetMap (OSM). Accommodations (2,800) were gathered from booking platforms. Flights (3,446,829 records) were adapted from TravelPlanner (Xie et al., 2024) with date alignment to the event timeline. Events (21,980) were scraped from web sources covering four types: Sports, Arts & Theatre, Music, and Film. Public transit data (8,723 nearest-transit-stop entries) were obtained from GTFS feeds for all 140 cities. A distance matrix with 19,460 entries supports spatial scoring.

City selection and difficulty. Cities were filtered to those covered by GTFS data. Difficulty levels reflect data density: Easy cities have abundant PoI, restaurant, and event data; Hard cities are sparsely covered (Table 2). For city coverage per trip, 3-day itineraries involve one city, 5-day involve two, and 7-day involve three.

Days	Easy	Medium	Hard	Total
3-day	130	106	108	344
5-day	87	164	73	324
7-day	99	178	55	332

Table 2: *TripCraft* dataset distribution

Cleaning and alignment. Missing entries were removed or replaced with sensible defaults. Dates were synchronized to the event timeline to ensure that event references in itineraries correspond to real occurrences. Geographic consistency across PoIs, accommodations, and transit stops was verified programmatically.

Query and persona generation. Departure city, destination(s), and travel dates were randomized within coverage constraints. User personas were sampled from four components: Traveler Type, Purpose of Travel, Spending Preference, and Location Preference (Table 3). Constraints and personas were injected into GPT-4o few-shot prompts to generate high-quality natural-language queries. Fifteen attraction categories and four event types were used as hard constraints, while meal-gap rules and PoI validity rules formed commonsense constraints.

Annotation protocol. Twenty-five graduate annotators produced gold itineraries with written rationales justifying each PoI selection, meal choice, and accommodation. The annotation process involved iterative refinement with expert feedback; each instance required approximately 30 minutes. Final review combined automated script checks and manual expert adjudication to ensure consistency, temporal feasibility, and persona alignment.

3.3 Constraints and Personas

TripCraft enforces both commonsense and hard constraints. Commonsense constraints include: (1) Sufficient Meal Gaps (minimum four hours between meals), (2) Valid PoI list (PoIs must be real entries in the database), and (3) Diverse Events (no repeated event choices within a trip). Hard constraints specify that all attractions must belong to one of 15 predefined categories and all events to one of four types. User personas inject traveler preference signals (Table 3).

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Commonsense Constraints	
Sufficient Meal Gaps	Defines a minimum gap of four hours between the meals.
Valid PoI list	Defines validity rules for the Point of Interest list.
Diverse Events	Event choices should not be repeated throughout the trip.
Hard Constraints	
Event Types	Event Types include four distinct categories—Sports, Arts & Theatre, Music, and Film.
Attraction Types	Each attraction belongs to one or more of 15 predefined categories, ensuring a well-distributed selection of activities.
Persona Components	
Traveler Type	Defines how a traveler approaches their journey—whether they seek relaxation in cozy spots or adrenaline-pumping adventures.
Purpose of Travel	Captures trip motivation. Examples: to unwind, explore cultures etc.
Spending Preference	Reflects the traveler's budget and style, from luxurious indulgence to cost-conscious experiences.
Location Preference	Highlights preferred environments, such as beaches, mountains, cities, or wildlife-rich forests.

Table 3: Addition of constraints and persona details based on availability and limitations of scraped data has been given in the above table. The full list of constraints used in *TripCraft* has been given in Table 9 of Appendix.

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3.4 Attraction Categories and Visit Durations

Each attraction is assigned a category, and each category carries a characteristic visit duration used in temporal scoring. Table 4 lists a representative subset; the full list contains 15 categories. Durations range from 2.5 hours for Nightlife to 4.5 hours for Nature & Parks, reflecting empirical norms that inform whether an LLM-generated plan allocates realistic time at each venue.

4 Evaluation Metrics

4.1 Traditional Constraint Metrics

Following TravelPlanner (Xie et al., 2024), we compute: **Delivery Rate** (fraction of queries for which a parseable plan was produced), **Commonsense Pass Rate (CPR)** measured micro and macro across all commonsense constraints, **Hard Constraint Pass Rate (HCPR)** measured micro and macro, and **Final Pass Rate** (all constraints simultaneously satisfied).

Category	Duration (hrs)
Boat Tours & Water Sports	3.5
Casinos & Gambling	2.5
Museums	3.0
Nature & Parks	4.5
Nightlife	2.5
Sights & Landmarks	3.0

Table 4: Attraction visiting duration (hrs) for a subset of categories (due to page limit). The complete list of categories has been given in Table 8 of Appendix.

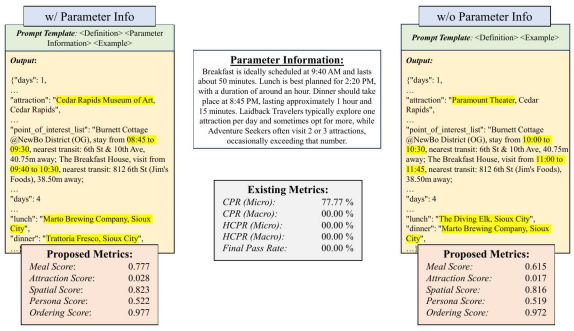

Figure 3: Existing constraint-based metrics rate these two plans equally, but our continuous metrics reveal differences in temporal scheduling, spatial transit proximity, PoI ordering, and persona alignment.

4.2 Novel Continuous Metrics

Binary pass rates cannot distinguish a near-perfect plan from a fundamentally poor one along quality dimensions that matter in practice. Figure 3 illustrates how two plans that receive identical binary scores differ substantially under our continuous metrics. *TripCraft* introduces five complementary scores.

Temporal Meal Score ($\bar{T}_{meal}$). For each meal (breakfast, lunch, dinner), we compute the midpoint time and duration from the generated itinerary and evaluate them under a bivariate Gaussian dis-

tribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ fitted to the annotation distribution, where $\boldsymbol{\mu} = [\mu_t, \mu_d]^T$ encodes mean meal time and duration, and $\boldsymbol{\Sigma}$ encodes their variances and covariance. The score is normalized by the maximum PDF value of the distribution. This rewards plans that schedule meals at realistic times with realistic durations and penalizes outliers (e.g., dinner at 3 p.m.).

Temporal Attraction Score ($\bar{T}_{attrac}$). For each attraction, we evaluate the visited duration against a per-category Gaussian centered at the expected duration μ_d, adjusted for traveler pace. For adventure seekers, the expected duration decreases with the number of attractions per day (more attractions $\Rightarrow$ shorter stays per venue), governed by Poisson count probability with rate $\lambda_{adventurous}$. For laidback travelers, the expected duration increases with fewer attractions per day, using $\lambda_{laidback}$. Parameters are shown in Table 5.

Spatial Score ($\bar{S}_{spatial}$). For each PoI in the plan, we extract the distance to the nearest public transit stop (from the GTFS-derived database). A piecewise score function penalizes large distances: distances up to 5,000 m are scored linearly from 1.0 to 0.5, and distances beyond 5,000 m are penalized exponentially. Day-level scores are averaged over PoIs; the final score averages over all days. This rewards plans that are transit-accessible and penalizes itineraries that strand travelers far from public transport.

Ordering Score ($\bar{S}_{ord}$). The ordering score measures how closely the generated PoI sequence matches the gold-annotated sequence in terms of visit type ordering (restaurant y, accommodation x, attraction z). We compute a Weighted Edit Distance (WED) between the generated and gold sequences using the Levenshtein distance formulation (Levenshtein, 1965), then normalize by sequence length and invert to obtain a similarity score. Intuitively, this rewards plans that visit attractions during the day, meals at appropriate intervals, and accommodations at night, in a realistic order.

Persona Score ($\bar{S}_{persona}$). For each PoI in the plan, we compute its BERT (Devlin et al., 2019) embedding and average the cosine similarities with the embeddings of each persona component (Traveler Type, Purpose of Travel, Spending Preference, Location Preference). The final score averages over all PoIs and persona components. This rewards plans that select venues semantically aligned with the user's persona.

Parameter	3-day	5-day	7-day
Restaurant Parameters			
Breakfast			
Mean Duration (hrs)	0.90	1.08	0.85
Std. Duration (hrs)	0.24	1.43	0.23
Lunch			
Mean Duration (hrs)	1.11	1.10	0.99
Std. Duration (hrs)	0.36	0.35	0.26
Dinner			
Mean Duration (hrs)	1.19	1.32	1.15
Std. Duration (hrs)	0.43	0.91	1.15
Attraction Parameters			
$\lambda_{laidback}$	1.10	1.26	1.11
$\lambda_{adventurous}$	2.01	1.61	1.82
σ_d (hrs)	1.11	1.07	0.90
n^{max}	5	4	4
n^{min}	0	0	0
k (hrs)	0.28	0.28	0.28

Table 5: Day-wise parameter details calculated from our annotation distribution. For a complete list of parameters refer to the Appendix.

4.3 Parameter Derivation

All metric parameters are derived empirically from annotation statistics stratified by trip length and traveler type. Table 5 reports key parameters for the restaurant and attraction scoring components. For example, the adventurous Poisson rate $\lambda_{adventurous}$ varies from 1.82 (7-day) to 2.01 (3-day), reflecting that annotators packed more attractions per day in shorter trips for adventure-seeking personas.

5 Experiments and Results

5.1 Experimental Setup

We evaluate a GPT-4o-based travel planning agent on *TripCraft* under two settings: **(1) *w/o Parameter Info***: the agent receives the query, persona, and constraints but no information about the metric parameters; and **(2) *w/ Parameter Info***: the agent additionally receives the benchmark parameters (e.g., typical meal timing, expected attraction visit durations, per-persona pacing guidelines) as part of the prompt. This ablation tests whether grounding the agent in annotation-derived statistics improves itinerary quality.

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5.2 Results on Novel Continuous Metrics

Table 6 reports results on the five continuous metrics. Parameter-informed prompting yields consistent improvements across most metrics and trip lengths.

Temporal Meal Score. The most pronounced gains occur in meal scheduling. The 7-day Temporal Meal Score improves from 0.61 to 0.80 (+31%), and the 3-day score improves from 0.56 to 0.70 (+25%). This indicates that without explicit parameter guidance, LLMs schedule meals at times that deviate substantially from natural human meal patterns; providing the annotation-derived timing distributions substantially corrects this.

Temporal Attraction Score. The 3-day and 5-day Temporal Attraction Scores improve slightly with parameter information (0.0169→0.0171 and 0.0057→0.0078, respectively). Notably, the 7-day score decreases (0.0122→0.0057), suggesting that for longer trips, providing pacing parameters may cause the model to over-constrain its attraction scheduling in ways that deviate from the gold distribution. The raw values of $\bar{T}_{attrac}$ are small because they represent unnormalized PDF values; relative changes are the meaningful signal.

Spatial Score. Spatial scores are consistently high across both settings (0.80–0.87), indicating that GPT-4o already selects transit-proximate venues without explicit guidance. Parameter information yields small but consistent improvements (e.g., 3-day: 0.80→0.83).

Ordering Score. Ordering improves across all trip lengths with parameter information (3-day: 0.70→0.74; 5-day: 0.91→0.93; 7-day: 0.96→0.97). The baseline is already strong for longer trips, suggesting LLMs naturally structure multi-day itineraries with plausible visit ordering.

Persona Score. Persona alignment shows modest improvement (3-day: 0.46→0.48; 5-day: 0.49→0.50; 7-day: 0.50→0.51). The relatively low baseline values indicate that aligning PoI selections semantically with user personas remains a significant challenge for current LLMs.

5.3 Results on Traditional Constraint Metrics

Table 7 reports results on traditional constraint metrics. The picture here is more nuanced and reveals an important trade-off.

Delivery Rate. Parameter-informed prompting improves delivery for 3-day itineraries (92.60→96.08) but decreases it for 5-day (98.69→91.30) and 7-day (96.03→92.51). This suggests that for longer trips, additional prompt constraints make plan generation more brittle, occasionally causing the agent to fail to produce a parseable output.

Commonsense Pass Rate (CPR). For 3-day itineraries, parameter information slightly improves CPR Micro (77.61→80.08). For 5-day and 7-day itineraries, CPR Micro decreases (66.99→61.69 and 70.35→66.43, respectively). This suggests a rigidity vs. feasibility tension: constraining the agent more tightly to realistic temporal and spatial patterns may cause it to violate certain commonsense rules it would otherwise satisfy (e.g., meal gap requirements) as it attempts to optimize multiple objectives simultaneously.

Hard Constraint Pass Rate (HCPR). For 5-day itineraries, HCPR Micro improves with parameter information (4.21→8.60), while 3-day HCPR Micro decreases slightly (27.63→26.10). The 7-day HCPR Micro improves from 0.62 to 0.83, while both settings achieve 0.00 on HCPR Macro, indicating that fully satisfying all hard constraints simultaneously remains very difficult for multi-day itineraries.

Final Pass Rate. Final pass rates are uniformly low across all settings, confirming TravelPlanner's observation that current LLMs struggle with simultaneous multi-constraint satisfaction at scale (Xie et al., 2024). Parameter-informed prompting generally decreases final pass rates, reinforcing the trade-off between continuous quality improvement and strict binary constraint satisfaction.

5.4 Discussion

The results expose a fundamental tension in LLM travel planning evaluation: optimizing for continuous quality dimensions (realistic temporal scheduling, persona fit) can conflict with rigid binary constraint satisfaction. *TripCraft*'s continuous metrics capture a dimension of plan quality that binary pass rates miss entirely. A plan that satisfies all hard constraints but schedules dinner at 3 p.m. for an adventure seeker craving nightlife is, in practical terms, a poor plan. *TripCraft*'s metrics make these distinctions measurable.

The low persona scores (0.46–0.51) across both

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Setting	Category	$\bar{T}_{meal}$	$\bar{T}_{attrac}$	$\bar{S}_{spatial}$	$\bar{S}_{persona}$	$\bar{S}_{ord}$
w/o Parameter Info	3-day	0.56	0.0169	0.80	0.46	0.70
	5-day	0.69	0.0057	0.86	0.49	0.91
	7-day	0.61	**0.0122**	0.85	0.50	0.96
w/ Parameter Info	3-day	**0.70**	**0.0171**	**0.83**	**0.48**	**0.74**
	5-day	**0.75**	**0.0078**	**0.87**	**0.50**	**0.93**
	7-day	**0.80**	0.0057	**0.86**	**0.51**	**0.97**

Table 6: Results of our proposed evaluation metrics for the two settings. For each category, the **bolded** values represent the better ones between the two settings.

Setting	Category	Delivery Rate	CPR		HCPR		Final Pass Rate
			Micro	Macro	Micro	Macro	
w/o Parameter Info	3-day	92.60	77.61	**6.08**	**27.63**	**26.08**	**3.47**
	5-day	**98.69**	**66.99**	**2.17**	4.21	3.04	**1.74**
	7-day	**96.03**	**70.35**	0.00	0.62	0.00	0.00
w/ Parameter Info	3-day	**96.08**	**80.08**	5.21	26.10	25.21	1.74
	5-day	91.30	61.69	0.87	**8.60**	**5.21**	0.43
	7-day	92.51	66.43	0.00	**0.83**	0.00	0.00

Table 7: Results of existing evaluation metrics for the two settings. CPR and HCPR stand for Commonsense Pass Rate and Hard Constraint Pass Rate respectively. For each category, the **bolded** values represent the better ones between the two settings. Moreover, equal values have been underlined.

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settings reveal that LLMs have substantial room for improvement in aligning PoI selections with user personas. This points to an important direction for future work: developing persona-grounded planning strategies beyond surface-level prompt conditioning.

The consistently high spatial scores (0.80–0.87) deserve closer inspection. GPT-4o appears to exhibit a strong prior toward selecting well-known, centrally located venues that happen to be near transit hubs, which inflates spatial scores across the board. This effect likely masks performance differences on Hard-difficulty cities, where transit coverage is sparser and the model's urban-center bias becomes a more serious liability. Disaggregating spatial scores by city difficulty level would expose whether high aggregate scores reflect genuine transit awareness or merely a tendency to recommend popular downtown attractions.

Furthermore, the human-annotated gold itineraries constitute a resource beyond evaluation: the rationales written by 25 annotators encode implicit planning heuristics—such as day-opening attraction choices, intra-day neighborhood clustering, and meal venue sequencing relative to accommodation—that could serve as high-quality demonstrations for few-shot or retrieval-augmented planning approaches. We expect these gold plans to prove useful for training and prompting future LLM-based travel planners.

Despite these contributions, *TripCraft* has several limitations worth acknowledging. The benchmark is currently restricted to U.S. cities, limiting its geographic diversity; extending coverage to international destinations with varied transit systems, cultural meal norms, and distinct attraction vocabularies remains an important future direction. Additionally, gold itineraries inevitably reflect individual annotator styles—our 25-annotator pool, while carefully trained, may not represent the full diversity of real-world traveler expectations, and crowdsourced or larger-scale annotation could yield more varied reference plans. Finally, the Persona Score relies on BERT-based cosine similarity as a proxy for preference alignment, which may fail to capture nuanced signals such as budget sensitivity, dietary restrictions, or physical accessibility needs; future work could explore LLM-based evaluators or specialized preference models for richer persona-aware scoring.

6 Conclusion

We presented *TripCraft*, a spatio-temporally fine-grained travel planning benchmark that combines real-world, constraint-rich data with continuous, multi-faceted evaluation metrics. By grounding itinerary generation in GTFS public transit data,

live event calendars, typed attraction categories, and rich user personas—and by parameterizing evaluation metrics from human-annotated gold plans—*TripCraft* captures quality dimensions that existing binary evaluations miss. Our experiments show that parameter-informed prompting substantially improves temporal scheduling and spatial coherence while exposing a trade-off with strict constraint satisfaction, underscoring the practical value of continuous metrics for assessing LLM travel planners. We release *TripCraft* to facilitate future research on realistic, persona-conditioned travel planning.

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